A Minor Project Synopsis

On

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Secure Cloud Data Access Using Zero Trust Security Model

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Secure Cloud Data Access using Zero Trust Security Model

1. ABSTRACT

Cloud computing has become the backbone of modern digital infrastructure, enabling scalable, on-demand storage and computing services for enterprises and individuals. However, traditional security models based on perimeter defence assume that users within the network are trustworthy, which exposes systems to insider threats, credential theft, and unauthorised access.

This project proposes a **Secure Cloud Data Access System based on the Zero Trust Security Model**, which follows the principle *"Never Trust, Always Verify."* The system enforces strict identity verification, device validation, and dynamic access control before granting access to cloud resources.

The proposed solution integrates **Multi-Factor Authentication (MFA), Role-Based Access Control (RBAC), Risk-Based Adaptive Access Control, and Continuous Monitoring mechanisms** to ensure robust security. Additionally, rule-based risk evaluation using predefined conditions such as device, location, and login time", IP address, and login patterns.

The system is designed using modern technologies such as **Python, Flask, JWT authentication, and cloud platforms (AWS/Azure)** to create a scalable and real-time secure environment. The implementation ensures reduced unauthorised access, improved data protection, and enhanced system reliability.

2. INTRODUCTION

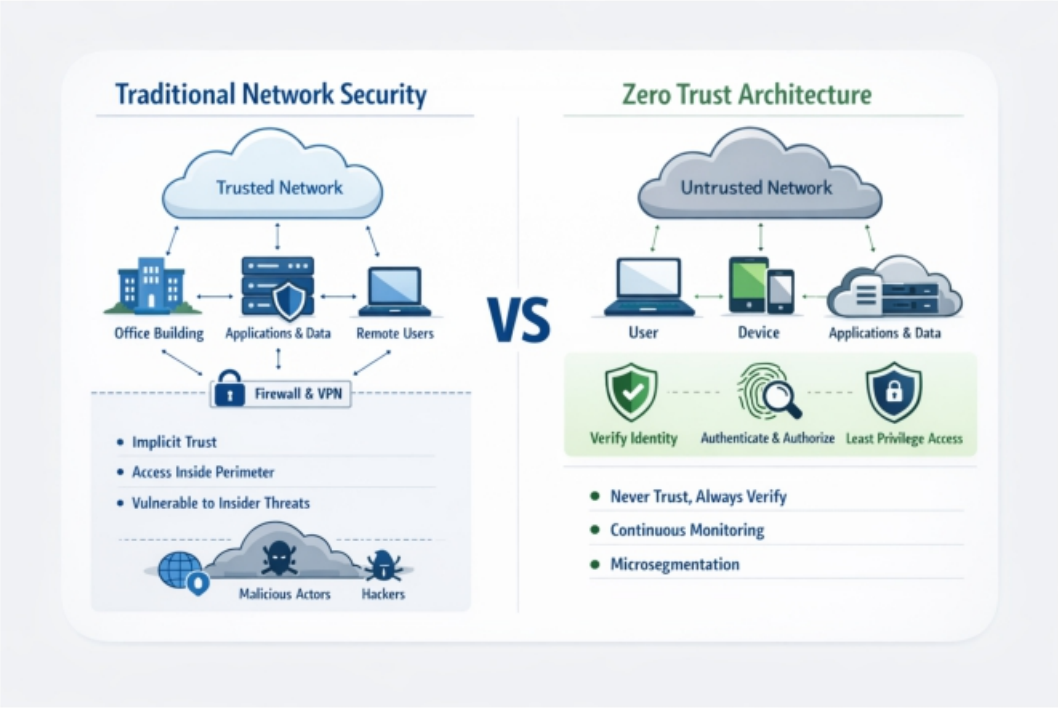
Cloud computing has revolutionised how organisations manage and access data. Services such as SaaS, PaaS, and IaaS allow users to access data remotely without maintaining physical infrastructure. However, with increasing adoption, cloud environments have become prime targets for cyberattacks.

Traditional security models rely on a **perimeter-based approach**, where users inside the network are trusted. This model fails in modern distributed systems where users access cloud services from multiple devices and locations.

To overcome these limitations, the **Zero Trust Security Model** has emerged as a modern approach. It eliminates the concept of implicit trust and enforces strict verification for every access request.

This project implements a **Zero Trust-based cloud security system** that ensures:

· Continuous authentication
· Context-aware authorisation
· Real-time monitoring
· Dynamic access decisions



Traditional vs Zero Trust Architecture Diagram

3. Problem Statement

Cloud computing is widely used for storing and accessing data, but it introduces significant security challenges. Traditional security models rely on **perimeter-based protection**, where users inside the network are trusted by default. This approach is ineffective in modern environments where users access cloud systems from multiple devices and locations.

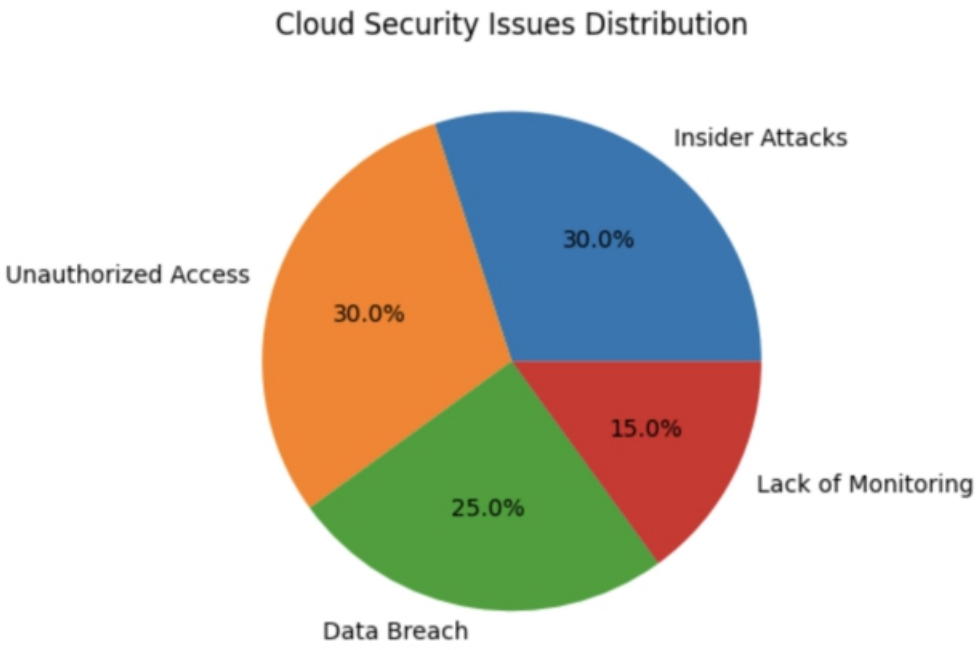
According to the analysis shown in the **Cloud Security Issues Pie Chart**, major threats include **Insider Attacks (30%)**, **Unauthorized Access (30%)**, **Data Breaches (25%)**, and **Lack of Monitoring (15%)**. These issues arise due to implicit trust, weak authentication mechanisms, and absence of continuous verification.

Insider attacks occur when trusted users misuse access privileges, while unauthorized access happens due to compromised credentials. Data breaches result from poor access control, and lack of monitoring prevents detection of malicious activities in real-time.

Traditional systems also fail to consider contextual factors such as user behaviour, device identity, and location, making them vulnerable to dynamic threats. Static access control mechanisms like basic RBAC are not sufficient to handle modern cloud security requirements.

Therefore, there is a need for a robust security framework based on the **Zero Trust Model**, which eliminates implicit trust and ensures continuous authentication, authorization, and monitoring. The proposed system follows the principle:

“Never Trust, Always Verify”, ensuring secure and reliable cloud data access.



Cloud Security Issues (Pie Chart)

4. OBJECTIVE

· To design a Zero Trust-based cloud security architecture
· To implement Multi-Factor Authentication (MFA)
· To apply Role-Based Access Control (RBAC)
· To integrate Rule-Based Risk Evaluation
· To enable Basic Session Validation and Logging and session validation
· To enhance overall cloud data security

5. LITERATURE REVIEW

5.1 Traditional Security Models

Traditional models rely on firewalls and network boundaries. These models assume that internal users are trustworthy, which is not suitable for modern cloud systems.

5.2 Zero Trust Architecture

NIST SP 800-207 defines Zero Trust as a model that removes implicit trust and enforces strict identity verification. Google's BeyondCorp is a real-world implementation.

5.3 Role-Based Access Control (RBAC)

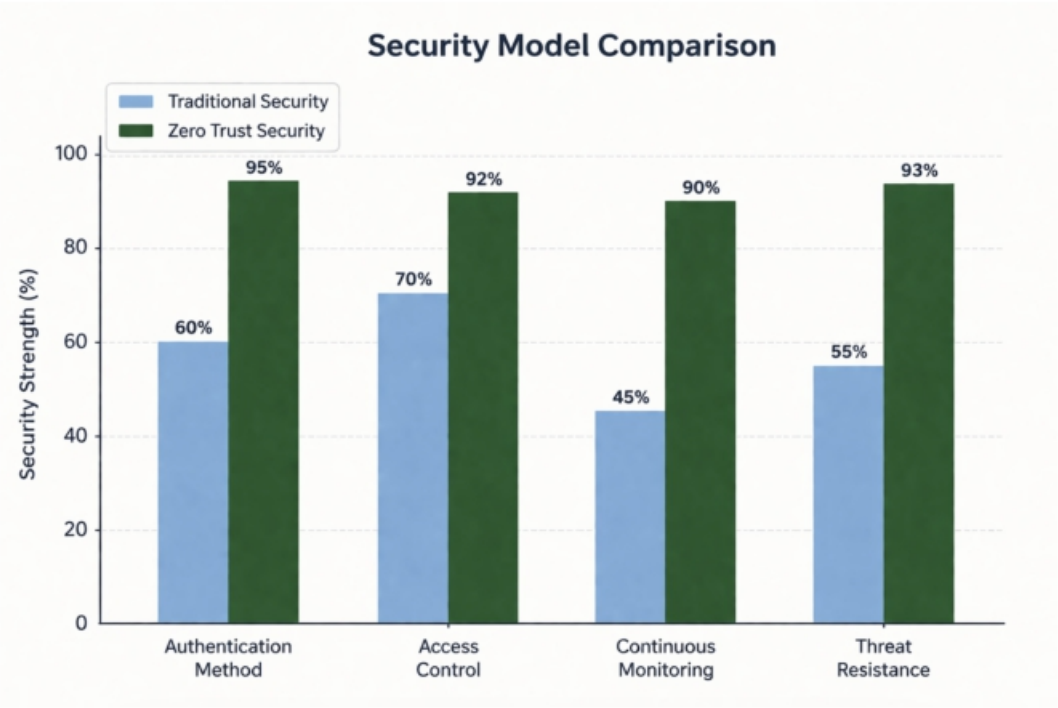
RBAC restricts access based on user roles. While effective, it lacks dynamic adaptability.

5.4 Risk-Based Access Control

Modern systems use adaptive security where access decisions depend on user behavior, location, and device.

Research Gap

Most systems either provide static access control or basic authentication. This project combines Zero Trust, RBAC, and rule-based risk evaluation in a practical implementation.



Security Model Comparison (Bar Graph)

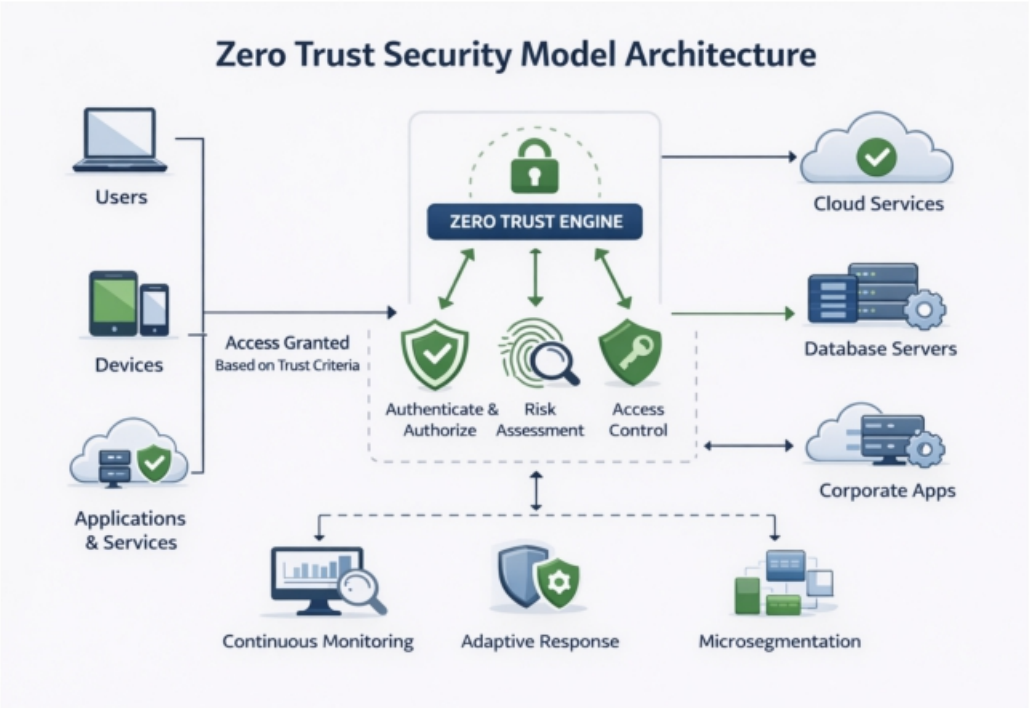
6. SYSTEM ARCHITECTURE

The proposed system consists of the following components:

· User Interface (Login & Access Portal)
· Authentication Server
· Database (Users, Roles, Logs)
· Frontend (Login & Dashboard)
· Backend (Authentication + Access Control API)
· Cloud Data Storage

Flow:
· User sends access request
· Authentication is performed using MFA
· Device and IP verification is done
· Policy decision is evaluated using RBAC
· Risk score is calculated

· Access is granted or denied
· Continuous monitoring is applied



Zero Trust System Architecture

7. Methodology

The system follows a structured approach:

Phase 1: Authentication

· Username and password verification
· OTP-based MFA

Phase 2: Device Verification

· Basic device recognition
· IP address validation

Phase 3: Access Control

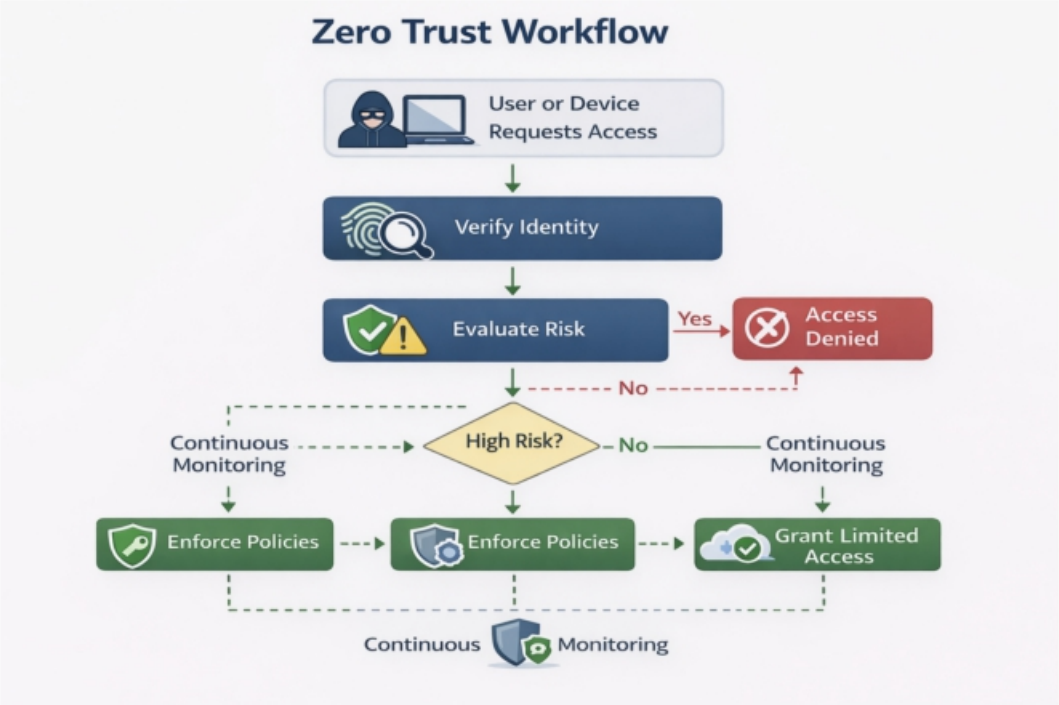
· Role-Based Access Control
· Least Privilege Principle

Phase 4: Risk Analysis

· Location tracking
· Predefined login conditions (time/location)

Phase 5: Continuous Monitoring

· Session tracking
· Session expiry
· Re-authentication if required



Zero Trust Workflow (Flowchart).

8. Algorithms Used

8.1 Authentication Algorithm (MFA)

· User enters credentials
· System verifies username and password
· OTP is generated
· OTP is verified
· Access granted or denied

8.2 RBAC Algorithm

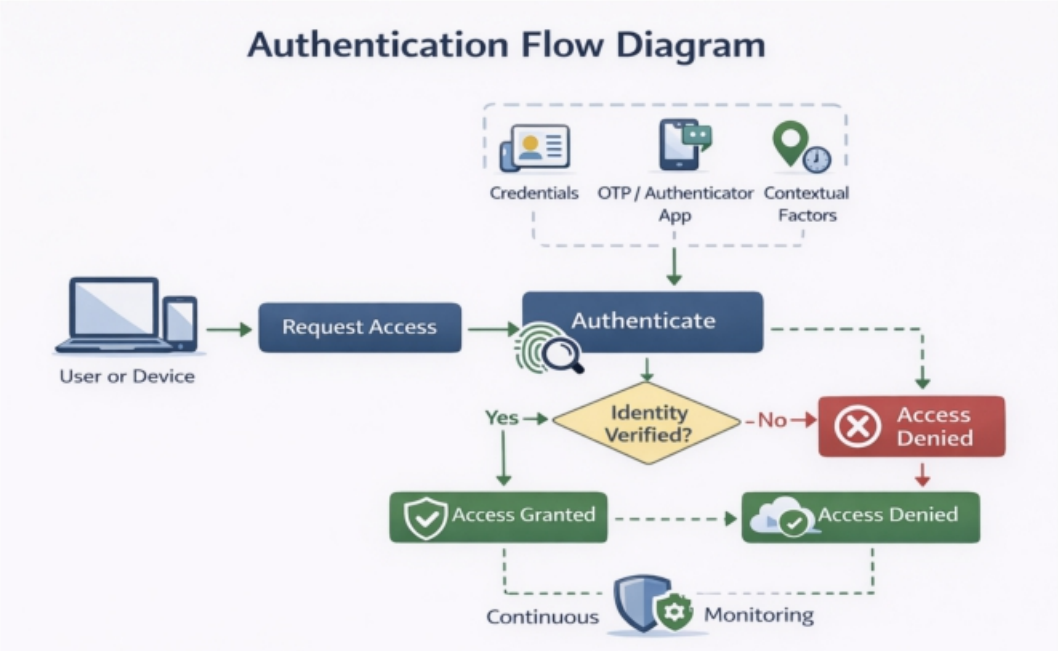
· Assign roles to users
· Map roles to permissions
· Match request with role
· Allow or deny access

8.3 Risk-Based Algorithm

· Collect user data (location, device, time)
· Calculate risk score
· Compare with threshold
· Allow, deny, or re-authenticate

8.4 Continuous Monitoring Algorithm

· Monitor user session
· Detect anomalies
· Trigger re-authentication
· Terminate session if required



Authentication Flow Diagram

9. Tools and Technologies

· **Programming Languages:** Python
· **Frameworks:** Flask / FastAPI
· **Database Systems:** MySQL / PostgreSQL
· **Cloud Platforms:** AWS, Azure
· **Security Mechanisms:** OAuth 2.0, JSON Web Token (JWT), Multi-Factor Authentication (MFA)

10. Expected Results

Working prototype demonstrating MFA + RBAC + risk-based access

The system provides the following improvements:

· Increased security level
· Reduced unauthorised access
· Improved monitoring
· Better access control

Security Improvement (Bar Graph) AND Figure 8: Unauthorised Access Reduction (Line Graph)

11. Advantages

· Eliminates the concept of implicit trust within the network
· Mitigates risks associated with insider threats
· Provides high scalability for modern cloud environments
· Ensures strong data protection and secure access control

12. Applications

· Banking systems
· Healthcare systems
· Government data security
· Enterprise cloud platforms

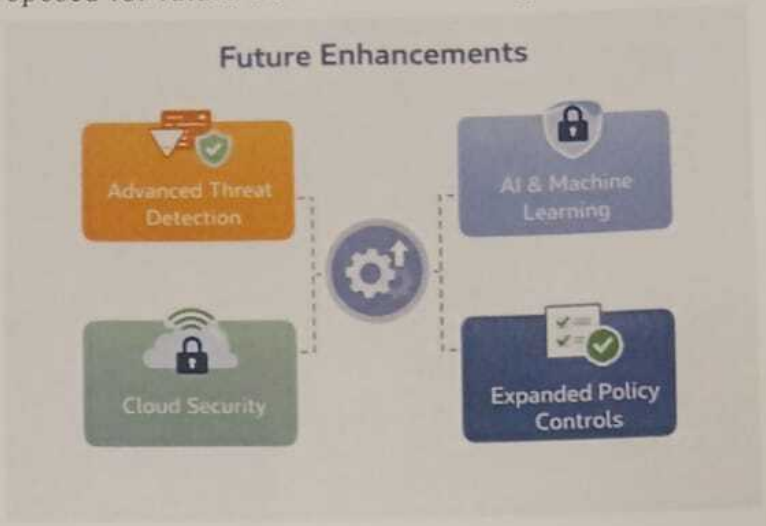
13. Conclusion

The Zero Trust Security Model ensures strong protection by eliminating implicit trust and enforcing continuous verification. It significantly reduces security risks and enhances cloud data protection.

14. Future Scope

- Integration with AI/ML for smart threat detection
- Biometric authentication
- Blockchain-based security enhancement
- Advanced Session expiry systems

These features are proposed for future work and are not implemented in this project



Future Enhancements Diagram

15. References (~~Sample~~)

- NIST SP 800-207 (Zero Trust Architecture)
- Google BeyondCorp Model

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